Difficulty Key: ★ = Basic/Straightforward ★★ = Moderate ★★★ = Challenging

1 Linear Algebra Review

Problem 1. (★ VMLS 1.10 Total score.) The record for each student in a class is given as a 10-vector r, where $r_1, \ldots, r_8$ are the grades for the 8 homework assignments, each on a $0-10$ scale, r_9 is the midterm exam grade on a $0-120$ scale, and r_{10} is the final exam score on a $0-160$ scale. The student's total course score s, on a $0-100$ scale, is based 25% on the homework, 35% on the midterm exam, and 40% on the final exam. Express s in the form $s = w^T r$. You can give the coefficients of w to 4 digits after the decimal point.

Solution. Let's first compute each component score on a 0-100 scale.
Step 1: Scale each component to 0-100.

- Total Homework Score (scaled): Since each HW is worth 10 points and there are 8 HWs, the total raw HW score is out of 80. To scale to 0-100:

$$\text{Total\_HW\_Score} = \frac{100}{80}(r_1 + r_2 + \cdots + r_8) = 1.25(r_1 + r_2 + \cdots + r_8)$$

- Midterm Score (scaled): The midterm is out of 120, so on a 0-100 scale:

$$\text{Midterm\_Score} = \frac{100}{120} r_9 = 0.8333 r_9$$

- Final Exam Score (scaled): The final is out of 160, so on a 0-100 scale:

$$\text{Final\_Score} = \frac{100}{160} r_{10} = 0.625 r_{10}$$

Step 2: Apply the weights to get the total score.
The total course score s is a weighted combination: 25% homework, 35% midterm, and 40% final.

$$\begin{aligned} s &= 0.25 \cdot \text{Total\_HW\_Score} + 0.35 \cdot \text{Midterm\_Score} + 0.40 \cdot \text{Final\_Score} \\ &= 0.25 \cdot 1.25(r_1 + \cdots + r_8) + 0.35 \cdot 0.8333 r_9 + 0.40 \cdot 0.625 r_{10} \\ &= 0.3125(r_1 + \cdots + r_8) + 0.2917 r_9 + 0.25 r_{10} \end{aligned}$$

Step 3: Express as an inner product $s = w^T r$.
Writing this as an inner product:

$$s = (0.3125, 0.3125, \ldots, 0.3125, 0.2917, 0.25)^T r$$

where the first 8 entries are all 0.3125 (one for each homework).
Therefore, $w = (0.3125\mathbf{1}_8, 0.2917, 0.2500)$, where $\mathbf{1}_8$ denotes an 8-dimensional vector of ones.

Problem 2. (★★ Linear independence of stacked vectors.) Consider the stacked vectors

$$c_1 = \begin{bmatrix} a_1 \\ b_1 \end{bmatrix}, \quad \ldots, \quad c_k = \begin{bmatrix} a_k \\ b_k \end{bmatrix},$$

where $a_1, \ldots, a_k$ are n-vectors and $b_1, \ldots, b_k$ are m-vectors.

(a) Suppose $a_1, \ldots, a_k$ are linearly independent. (We make no assumptions about the vectors $b_1, \ldots, b_k$.) Can we conclude that the stacked vectors $c_1, \ldots, c_k$ are linearly independent?

(b) Now suppose that $a_1, \ldots, a_k$ are linearly dependent. (Again, with no assumptions about $b_1, \ldots, b_k$.) Can we conclude that the stacked vectors $c_1, \ldots, c_k$ are linearly dependent?

Solution. **(a) Yes.** Suppose $\alpha_1 c_1 + \alpha_2 c_2 + \cdots + \alpha_k c_k = 0$ for some scalars $\alpha_1, \ldots, \alpha_k$. Then

$$\alpha_1 \begin{bmatrix} a_1 \\ b_1 \end{bmatrix} + \alpha_2 \begin{bmatrix} a_2 \\ b_2 \end{bmatrix} + \cdots + \alpha_k \begin{bmatrix} a_k \\ b_k \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

This means

$$\begin{bmatrix} \alpha_1 a_1 + \alpha_2 a_2 + \cdots + \alpha_k a_k \\ \alpha_1 b_1 + \alpha_2 b_2 + \cdots + \alpha_k b_k \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

Looking at the top block, we have

$$\alpha_1 a_1 + \alpha_2 a_2 + \cdots + \alpha_k a_k = 0.$$

Since $a_1, \ldots, a_k$ are linearly independent, we must have $\alpha_1 = \alpha_2 = \cdots = \alpha_k = 0$. Therefore, the only linear combination of $c_1, \ldots, c_k$ that equals zero is the trivial one, so $c_1, \ldots, c_k$ are linearly independent.

(b) No. We cannot conclude this in general. Here is a counterexample: Let $n = m = 1$ and $k = 2$. Take

$$a_1 = 1, \quad a_2 = 2, \quad b_1 = 1, \quad b_2 = -2.$$

Then a_1, a_2 are linearly dependent (since $2a_1 - a_2 = 0$), but the stacked vectors are

$$c_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad c_2 = \begin{bmatrix} 2 \\ -2 \end{bmatrix}.$$

These are linearly independent because no nonzero scalar multiple of one equals the other (e.g., if $\alpha c_1 = c_2$, then $\alpha = 2$ from the first component but $\alpha = -2$ from the second component, which is a contradiction).

Intuition: When the a_i are dependent, the b_i can potentially ”rescue” the independence of the stacked vectors. The dependence in the top blocks can be compensated by appropriate structure in the bottom blocks.

Problem 3. (★★ VMLS 3.5 Other common norms.) A vector norm is any real-valued function f satisfying the following four properties:

1. **Nonnegative homogeneity:** $\|\alpha x\| = |\alpha| \|x\|$ for all scalars α and vectors x.
2. **Triangle inequality:** $\|x + y\| \leq \|x\| + \|y\|$ for all vectors x and y.
3. **Nonnegativity:** $\|x\| \geq 0$ for all vectors x.
4. **Definiteness:** $\|x\| = 0$ if and only if $x = 0$.

The 1-norm and ∞-norm for n-vectors are defined as

$$\|x\|_1 = |x_1| + \cdots + |x_n|\,, \quad \|x\|_\infty = \max\{|x_1|\,, \ldots, |x_n|\}\,.$$

Verify that the 1-norm and the ∞-norm satisfy the four norm properties listed above.
Solution. a) Homogeneity.

$$\begin{aligned}\|\beta x\|_1 &= |\beta x_1| + |\beta x_2| + \cdots + |\beta x_n| \\ &= |\beta|\,|x_1| + |\beta|\,|x_2| + \cdots + |\beta|\,|x_n| \\ &= |\beta|\left(|x_1| + |x_2| + \cdots + |x_n|\right) \\ &= |\beta|\|x\|_1\end{aligned}$$

(b) Triangle inequality.

$$\begin{aligned}\|x+y\|_1 &= |x_1+y_1| + |x_2+y_2| + \cdots + |x_n+y_n| \\ &\le |x_1| + |y_1| + |x_2| + |y_2| + \cdots + |x_n| + |y_n| \\ &= \|x\|_1 + \|y\|_1\end{aligned}$$

(c) Nonnegativity. Each term in $\|x\|_1 = |x_1| + \cdots + |x_n|$ is nonnegative.
(d) Definiteness. $\|x\|_1 = 0$ only if $|x_1| = \cdots = |x_n| = 0$.
Next the ∞-norm. (a) Homogeneity.

$$\begin{aligned}\|\beta x\|_\infty &= \max\{|\beta x_1|\,, |\beta x_2|\,, \ldots, |\beta x_n|\} \\ &= \max\{|\beta|\,|x_1|\,, |\beta|\,|x_2|\,, \ldots, |\beta|\,|x_n|\} \\ &= |\beta|\max\{|x_1|\,, \ldots, |x_n|\} \\ &= |\beta|\|x\|_\infty\end{aligned}$$

(b) Triangle inequality.

$$\begin{aligned}\|x+y\|_\infty &= \max\{|x_1+y_1|\,, |x_2+y_2|\,, \ldots, |x_n+y_n|\} \\ &\le \max\{|x_1| + |y_1|\,, |x_2| + |y_2|\,, \ldots, |x_n| + |y_n|\} \\ &\le \max\{|x_1|\,, |x_2|\,, \ldots, |x_n|\} + \max\{|y_1|\,, |y_2|\,, \ldots, |y_n|\} \\ &= \|x\|_\infty + \|y\|_\infty\end{aligned}$$

(c) Nonnegativity. $\|x\|_\infty$ is the largest of n nonnegative numbers $|x_k|$.
(d) Definiteness. $\|x\|_\infty = 0$ only if $|x_k| = 0$ for $k = 1, \ldots, n$.

2 K-Means

Problem 4. (★★★ Minimizing mean square distance to a set of vectors.) Let $x_1, \ldots, x_L$ be a collection of n-vectors. Show that the vector z which minimizes the sum-square distance

$$J(z) = \|x_1 - z\|^2 + \cdots + \|x_L - z\|^2$$

is the centroid $\bar{x} = (1/L)(x_1 + \cdots + x_L)$.

(a) Explain why, for any z, we have

$$J(z) = \sum_{i=1}^{L} \|x_i - \bar{x} - (z - \bar{x})\|^2 = \sum_{i=1}^{L} \left(\|x_i - \bar{x}\|^2 - 2(x_i - \bar{x})^T(z - \bar{x})\right) + L\|z - \bar{x}\|^2.$$

(b) Explain why $\sum_{i=1}^{L}(x_i - \bar{x})^T(z - \bar{x}) = 0$. *Hint.* Write the left-hand side as

$$\left(\sum_{i=1}^{L}(x_i - \bar{x})\right)^T (z - \bar{x}),$$

and argue that the left-hand vector is 0.

(c) Combine the results of (a) and (b) to get $J(z) = \sum_{i=1}^{L} \|x_i - \bar{x}\|^2 + L\|z - \bar{x}\|^2$. Explain why for any $z \neq \bar{x}$, we have $J(z) > J(\bar{x})$. This shows that the choice $z = \bar{x}$ minimizes $J(z)$.

Solution. (a) Using the identity $\|a - b\|^2 = \|a\|^2 - 2a^T b + \|b\|^2$, we rewrite $x_i - z = (x_i - \bar{x}) - (z - \bar{x})$ and expand:

$$\|x_i - z\|^2 = \|x_i - \bar{x}\|^2 - 2(x_i - \bar{x})^T(z - \bar{x}) + \|z - \bar{x}\|^2.$$

Summing over $i = 1, \ldots, L$:

$$J(z) = \sum_{i=1}^{L} \|x_i - \bar{x}\|^2 - 2\sum_{i=1}^{L}(x_i - \bar{x})^T(z - \bar{x}) + L\|z - \bar{x}\|^2.$$

(b) We can factor out $(z - \bar{x})$ from the middle term:

$$\sum_{i=1}^{L}(x_i - \bar{x})^T(z - \bar{x}) = \left(\sum_{i=1}^{L}(x_i - \bar{x})\right)^T (z - \bar{x}).$$

Since $\bar{x} = \frac{1}{L}\sum_{i=1}^{L} x_i$, we have $\sum_{i=1}^{L}(x_i - \bar{x}) = \sum_{i=1}^{L} x_i - L\bar{x} = 0$. Thus the middle term vanishes.

(c) From parts (a) and (b):

$$J(z) = \sum_{i=1}^{L} \|x_i - \bar{x}\|^2 + L\|z - \bar{x}\|^2.$$

The first term is independent of z, and the second term is nonnegative, achieving its minimum value of 0 when $z = \bar{x}$. Therefore, $\bar{x}$ uniquely minimizes $J(z)$.

Problem 5. (★★★ Linear separation in 2-way partitioning.) Suppose we run k-means with $k = 2$ on the n-vectors $x_1, \ldots, x_N$, producing clusters with index sets G_1 and G_2. Show that there is a nonzero vector w and a scalar v that satisfy

$$w^T x_i + v \geq 0 \text{ for } i \in G_1, \qquad w^T x_i + v \leq 0 \text{ for } i \in G_2.$$

In other words, the affine function $f(x) = w^T x + v$ is greater than or equal to zero on the first group, and less than or equal to zero on the second group. This is called *linear separation* of the two groups (although *affine separation* would be more accurate).
Hint. Consider the function $\|x - z_1\|^2 - \|x - z_2\|^2$, where z_1 and z_2 are the group representatives.

Solution. Let z_1 and z_2 be the centroids of clusters G_1 and G_2, respectively. In k-means with $k = 2$, a point x_i is assigned to cluster 1 (i.e., $i \in G_1$) if and only if

$$\|x_i - z_1\|^2 \leq \|x_i - z_2\|^2,$$

and to cluster 2 (i.e., $i \in G_2$) if and only if

$$\|x_i - z_2\|^2 \leq \|x_i - z_1\|^2.$$

Following the hint, consider the function $g(x) = \|x - z_1\|^2 - \|x - z_2\|^2$. We expand this:

$$\begin{aligned}
g(x) &= \|x - z_1\|^2 - \|x - z_2\|^2 \\
&= (x - z_1)^T (x - z_1) - (x - z_2)^T (x - z_2) \\
&= x^T x - 2z_1^T x + z_1^T z_1 - x^T x + 2z_2^T x - z_2^T z_2 \\
&= 2(z_2 - z_1)^T x + (\|z_1\|^2 - \|z_2\|^2).
\end{aligned}$$

This is an affine function of x. Define

$$w = 2(z_2 - z_1), \qquad v = \|z_1\|^2 - \|z_2\|^2.$$

Then $g(x) = w^T x + v$, and we have:

- For $i \in G_1$: $\|x_i - z_1\|^2 \leq \|x_i - z_2\|^2$ implies $g(x_i) = \|x_i - z_1\|^2 - \|x_i - z_2\|^2 \leq 0$, so $w^T x_i + v \leq 0$.
- For $i \in G_2$: $\|x_i - z_2\|^2 \leq \|x_i - z_1\|^2$ implies $g(x_i) = \|x_i - z_1\|^2 - \|x_i - z_2\|^2 \geq 0$, so $w^T x_i + v \geq 0$.

Wait, this gives us the opposite signs! To match the problem statement, we should define

$$w = 2(z_1 - z_2), \qquad v = \|z_2\|^2 - \|z_1\|^2.$$

Then $f(x) = w^T x + v = -g(x) = \|x - z_2\|^2 - \|x - z_1\|^2$, and:

- For $i \in G_1$: $\|x_i - z_1\|^2 \leq \|x_i - z_2\|^2$ implies $f(x_i) = \|x_i - z_2\|^2 - \|x_i - z_1\|^2 \geq 0$, so $w^T x_i + v \geq 0$. ✓
- For $i \in G_2$: $\|x_i - z_2\|^2 \leq \|x_i - z_1\|^2$ implies $f(x_i) = \|x_i - z_2\|^2 - \|x_i - z_1\|^2 \leq 0$, so $w^T x_i + v \leq 0$. ✓

Finally, w is nonzero as long as $z_1 \neq z_2$, which holds whenever the two clusters are distinct.
Therefore, the affine function $f(x) = w^T x + v$ with $w = 2(z_1 - z_2)$ and $v = \|z_2\|^2 - \|z_1\|^2$ linearly separates the two clusters.

3 k-Nearest Neighbors

Problem 6. (★★★ Weighted Norm Decision Boundaries.) In weighted 1-NN classification, the decision boundary between two training points $a, b \in \mathbb{R}^d$ is the set of points x satisfying

$$d_W(x,a) = d_W(x,b), \quad \text{where} \quad d_W(x,y) = \sqrt{(x-y)^\top W(x-y)}, \quad W = \operatorname{diag}(\alpha_1, \dots, \alpha_d),\ \alpha_i > 0.$$

(a) Show that the decision boundary has the form of a hyperplane

$$c^\top x = d,$$

for some vector c and scalar d depending on a, b, W.

(b) Specialize your result to $d = 2$ and write the equation of the decision boundary in terms of $\mathbb{R}^2$coordinates x_1, x_2.

(c) Again with $d = 2$, show that if $\alpha_1 \neq \alpha_2$, this boundary is not perpendicular to the segment ab in the original coordinates.

(d) Again with $d = 2$, let $z = Sx$ with $S = \operatorname{diag}(\sqrt{\alpha_1}, \sqrt{\alpha_2})$. Show that in the z-coordinates the decision boundary is the usual perpendicular bisector of the transformed points Sa and Sb. Explain how the parameters (α_1, α_2) rescale the coordinate axes: if $\alpha_1 > \alpha_2$, distances along the x_1 direction are stretched more, so the first feature has greater influence in the weighted 1-NN classification, and similarly for $\alpha_2 > \alpha_1$.

Solution.

(a) Squaring both sides: $(x-a)^\top W(x-a) = (x-b)^\top W(x-b)$. Expanding and canceling $x^\top W x$:

$$-2x^\top W a + a^\top W a \;=\; -2x^\top W b + b^\top W b \quad\Longleftrightarrow\quad \boxed{(W(b-a))^\top x \;=\; \tfrac{1}{2}\big(\|b\|_W^2 - \|a\|_W^2\big)}$$

where $\|u\|_W^2 := u^\top W u$. This is a hyperplane with $c = W(b-a)$ and $d = \frac{1}{2}\big(\|b\|_W^2 - \|a\|_W^2\big)$.

(b) With $W = \operatorname{diag}(\alpha_1, \alpha_2)$ and $x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$, substituting into part (a) yields

$$\boxed{\alpha_1(b_1 - a_1)x_1 + \alpha_2(b_2 - a_2)x_2 = \tfrac{1}{2}\big[\alpha_1(b_1^2 - a_1^2) + \alpha_2(b_2^2 - a_2^2)\big]\,.}$$

Rearranging gives the weighted perpendicular bisector: $(b-a)^\top W\left(x - \frac{a+b}{2}\right) = 0$.

(c) The decision boundary from part (a) is

$$(W(b-a))^\top x = \tfrac{1}{2}\big(\|b\|_W^2 - \|a\|_W^2\big),$$

which is a hyperplane with normal vector

$$n = W(b-a).$$

The line segment connecting a and b has direction vector

$$v = b - a.$$

The boundary is *perpendicular* to the segment ab if and only if the normal n is parallel to v, that is,

$$W(b-a) = \lambda(b-a), \quad \text{for some } \lambda \in \mathbb{R}.$$

This requires that $b-a$ be an eigenvector of W. For

$$W = \mathrm{diag}(\alpha_1, \alpha_2),$$

the eigenvectors are the coordinate axes, so this condition holds only when $\alpha_1 = \alpha_2$ (the isotropic case) or when $b-a$ is aligned with one of the axes. Hence, for generic a, b and $\alpha_1 \neq \alpha_2$, the boundary is *not perpendicular* to the segment ab in the original coordinates.

(d) Let $S = \mathrm{diag}(\sqrt{\alpha_1}, \sqrt{\alpha_2})$ and $z = Sx$. Then $W = S^2$ and

$$d_W(x,a) = \|S(x-a)\|_2 = \|z - Sa\|_2.$$

Thus $d_W(x,a) = d_W(x,b)$ becomes $\|z - Sa\|_2 = \|z - Sb\|_2$, which is the Euclidean perpendicular bisector of Sa and Sb in z-coordinates.
Geometric intuition: S rescales axes via $z_i = \sqrt{\alpha_i}\, x_i$. If $\alpha_1 > \alpha_2$, the x_1-direction is stretched more in z-space, giving the first feature greater influence on nearest-neighbor decisions.

4 Least Squares

Problem 7. (★★ VMLS 12.1 Approximating a vector as a multiple of another one.) Find a scalar x that minimizes $\|ax - b\|^2$, where a and b are nonzero m-vectors. Show that $\|a\hat{x} - b\|^2 = \|b\|^2(\sin\theta)^2$, where $\theta = \angle(a, b)$.

Solution. For the least squares problem with $A = a$ (an m-vector), we want to minimize $\|ax - b\|^2$. The least squares solution is given by

$$\hat{x} = (a^T a)^{-1} a^T b = \frac{a^T b}{a^T a} = \frac{a^T b}{\|a\|^2}.$$

The residual is

$$a\hat{x} - b = a \cdot \frac{a^T b}{\|a\|^2} - b = \frac{a^T b}{\|a\|^2} a - b.$$

The squared norm of the residual is

$$\|a\hat{x} - b\|^2 = \left\| \frac{a^T b}{\|a\|^2} a - b \right\|^2.$$

To compute this, we expand:

$$\begin{aligned}
\|a\hat{x} - b\|^2 &= \left\| \frac{a^T b}{\|a\|^2} a \right\|^2 - 2 \left(\frac{a^T b}{\|a\|^2} a \right)^T b + \|b\|^2 \\
&= \frac{(a^T b)^2}{\|a\|^4} \|a\|^2 - 2 \frac{a^T b}{\|a\|^2} a^T b + \|b\|^2 \\
&= \frac{(a^T b)^2}{\|a\|^2} - 2 \frac{(a^T b)^2}{\|a\|^2} + \|b\|^2 \\
&= \|b\|^2 - \frac{(a^T b)^2}{\|a\|^2}.
\end{aligned}$$

Now, recall that $a^T b = \|a\|\|b\| \cos\theta$, where $\theta = \angle(a, b)$. Substituting:

$$\|a\hat{x} - b\|^2 = \|b\|^2 - \frac{(\|a\|\|b\| \cos\theta)^2}{\|a\|^2} = \|b\|^2 - \|b\|^2 \cos^2\theta = \|b\|^2(1 - \cos^2\theta).$$

Using the trigonometric identity $\sin^2\theta + \cos^2\theta = 1$, we get $1 - \cos^2\theta = \sin^2\theta$. Therefore,

$$\|a\hat{x} - b\|^2 = \|b\|^2 \sin^2\theta.$$

Interpretation: The optimal residual squared is $\|b\|^2 \sin^2\theta$. When $\theta = 0$ (i.e., a and b are parallel), the residual is zero, meaning b can be exactly represented as a multiple of a. When $\theta = 90^\circ$ (i.e., a and b are orthogonal), the residual is maximized at $\|b\|^2$, meaning a provides no useful approximation to b. The relative error is $\sin\theta$, which depends only on the angle between the vectors.

Problem 8. (★★ VMLS 12.7 Network Tomography.) A network has n links with unknown delays $d \in \mathbb{R}^n$. We measure travel times $t \in \mathbb{R}^N$ along N paths ($N > n$), where

$$P_{ij} = \begin{cases} 1 & \text{link } j \text{ is on path } i \\ 0 & \text{otherwise.} \end{cases}$$

Estimate the link delays $\hat{d}$ by minimizing the RMS deviation between measured times t and predicted times. Give a matrix expression for $\hat{d}$ and state any required assumptions.

Solution. The model for the travel times is

$$t \approx Pd,$$

where Pd is the predicted travel time vector. The i-th component of Pd is

$$(Pd)_i = \sum_{j=1}^{n} P_{ij} d_j = \sum_{j \text{ on path } i} d_j,$$

which is exactly the sum of link delays along path i.
To minimize the RMS deviation between measured times t and predicted times Pd, we minimize

$$\|Pd - t\|^2 = \|t - Pd\|^2.$$

This is the standard least squares problem. The least squares solution is given by

$$\hat{d} = (P^T P)^{-1} P^T t.$$

Assumptions required:

- The matrix $P^T P$ must be invertible. This is equivalent to requiring that P has *linearly independent columns* (i.e., $\text{rank}(P) = n$).
- In the network context, this means that the N paths must collectively provide enough information to uniquely determine all n link delays. For example, if a particular link never appears on any measured path, then its delay cannot be estimated.
- Alternatively, we can use the pseudo-inverse: $\hat{d} = P^\dagger t$, which works even when $P^T P$ is not invertible, giving the minimum-norm solution.

Physical interpretation: Each path measurement provides one linear equation relating the link delays. With $N > n$ measurements (overdetermined system), we have more equations than unknowns. The least squares solution $\hat{d}$ finds the link delays that best explain all the path measurements in the sense of minimizing the sum of squared errors.